

# Dinupa Devinda

Machine Learning Focused Graduate | AI/ML Developer | R&D Engineer  
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## PROFILE

Machine Learning-focused engineering graduate with a background in computer science, mathematics, Electrical and Electronics Engineering, and data analysis. Experienced in building practical ML systems, from data preparation and model development to evaluation and deployment focused implementation. Interested in developing intelligent, data driven solutions for real world technology and engineering problems.

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## EDUCATION

Sri Lanka Institute of Information Technology (SLIIT)  
**BSc. Engineering (Hons) in Electrical and Electronic Engineering**  
2022 – 2026 | Sri Lanka

University of Kelaniya  
**BSc. in Physical Science**  
(Subjects: Computer Science, Pure Mathematics, Applied Mathematics)  
2022 – 2026 | Sri Lanka

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## SKILLS & EXPERTISE

Programming Languages: Python | C/C++ | Java | JavaScript | TypeScript | C# | SQL

Frameworks & Tools: PyTorch | TensorFlow/Keras | scikit-learn | XGBoost | Optuna | NumPy | Pandas | OpenCV | YOLO | ONNX Runtime | TensorFlow Lite | NCNN | joblib | Cloudflare Workers | Cloudflare Vectorize | Chroma | Spring Boot | React | Next.js | MongoDB | SQL Server | Docker | Git | GitHub Actions | MQTT/MQTTS | ThingsBoard

Domains: Machine Learning | Deep Learning | RAG | LLM Applications | Embeddings | Vector Search | Computer Vision | Object Detection | Edge AI | Sensor Fusion | Audio Classification | Time-Series Classification | Feature Engineering | Model Evaluation | Data Analysis | Rest API Development

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## PROFESSIONAL EXPERIENCE

### Research and Development Engineering Intern

Sri Lanka Telecom – Digital Lab | Hybrid | June 2025 – September 2025

- Developed C/C++ firmware for ESP32-based edge devices with A7680C 4G modem AT-command handling, MQTT/MQTTS, HTTP, SMTP, SMS, and call support.
- Built device-to-cloud telemetry pipelines using MQTT/HTTP and structured JSON payloads for live monitoring, alerts, remote status reporting, and downstream analysis.
- Configured ThingsBoard dashboards and custom web portals, using MQTT Explorer, HiveMQ, and basic Grafana exposure for topic testing, JSON payload validation, telemetry debugging, and basic GitHub Actions, CI/CD workflow practice.

### Research and Development Inplant Trainee – Electronic Manufacturing Services Floor

Variosystems (Pvt) Ltd | Onsite | June 2024 – September 2024

- Built C/C++ based automation and embedded-control projects for a manufacturing environment, including an Automated Guided Vehicle, flame-treatment automation upgrade, and automated PCB conveyor system.
  - Gained hands-on exposure to production data flow, SMT/THT manufacturing, ICT and functional testing, SPEA 3030/4050 test systems, FactoryLogix workflows, and NPI processes.
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## PROJECTS

### **Vehicular Black Box: Vehicle Violation Detector**

Machine Learning, Edge AI, Sensor Fusion| Nov 2025 – May 2026

- Built and integrated a Raspberry Pi based vehicle violation detection system combining computer vision, audio classification, IMU-based ML, GPS/GSM communication, backend APIs, dashboard visualization, and driver scoring, with documented datasets, model cards, evaluation summaries, and a BYD Atto BMI160 IMU dataset published on [Zenodo/Kaggle](#).

**Road Sign Detection:** Developed a two-stage computer vision inference pipeline using YOLO, NCNN, ONNX Runtime, and OpenCV for Raspberry Pi deployment, evaluated on 9,062 test images.

**Horn Detection:** Built an audio classification model using log-mel spectrogram feature extraction, TensorFlow/Keras CNN training, and TensorFlow Lite deployment for live microphone inference.

**Crash Detection:** Developed crash-detection models using PyTorch CNN audio inference and TensorFlow/Keras CNN-GRU IMU time-series inference for multi-signal event detection.

**Lane Change Detection:** Built an IMU time-series classification model using accelerometer/gyroscope windowing, feature engineering, XGBoost training, and joblib deployment artifacts.

**Aggressive Driving Detection:** Developed an IMU-based machine learning classifier using yaw rate and raw acceleration windows, model comparison, Optuna-tuned XGBoost, threshold tuning, and evaluation.

### **AI Portfolio Assistant – Hybrid RAG Chatbot**

Next.js, Cloudflare Workers, Docker, Gemini API, Cloudflare Vectorize| Jul 2026

- Built and deployed a secure hybrid RAG portfolio assistant using Gemini API, Cloudflare Workers, and Cloudflare Vectorize, combining embedding-based vector search, keyword retrieval, reciprocal-rank fusion, validated source citations, lexical fallback, CORS controls, rate limiting, prompt-injection screening, protected reindexing, and a 20/20 offline retrieval evaluation; also Dockerized the local Python with Chroma RAG lab for reproducible chunking, embedding, indexing, testing, and evaluation workflows.

### **ServiceLink: On-Demand Local Hiring Platform**

Full-Stack Development, Spring Boot REST APIs| Oct 2025 – Jan 2026

- Built a full-stack service marketplace using Java 17, Spring Boot REST APIs, Spring Security/JWT authentication, MongoDB, React, Vite, React Router, React Query, and Axios, implementing role-based access, service listings, bookings, admin workflows, and API-driven frontend/backend communication.

### **A7680C 4G IoT Communication and Industrial Monitoring System**

C/C++, IoT Communication, MQTT/MQTTS, HTTP, ThingsBoard | June 2025 – Sep 2025

- Built an ESP32-based A7680C 4G communication library and industrial monitoring system supporting modem initialization, APN and network setup, MQTT/MQTTS over TLS, HTTP GET/POST, SMS, calls, SMTP alerts, TLS configuration, and web-based device setup, while collecting and validating machine data and streaming secure JSON telemetry to ThingsBoard and browser-based WebSocket MQTT dashboards for alerts, connection monitoring, and remote status visualization.

### **Other Projects**

- Built automation and monitoring projects including a Raspberry Pi based **Smart Fish Habitat Management System**, an Arduino based **Automated Guided Vehicle**, and an **automated PCB conveyor system**, applying Python, embedded C/C++, control logic, real-time monitoring, process automation, sensor data handling, and system integration. Completed additional projects including **BRMS Boarding Room Management API** with Java, Spring Boot REST APIs, Spring Data JPA, SQL Server, environment-based configuration, and H2-backed testing, Music Streaming Management System in Java, **Public Transport file-based database system in C**, **Music Player App** in C#/.NET, **EduPress Store** using PHP/MySQL/XAMPP.

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## CERTIFICATIONS

- PyTorch Bootcamp (OpenCV University, 2025)
- AI/ML Engineer Stage 2 (SLIIT, 2025)
- Machine Learning with Python Professional Certificate (LinkedIn Learning and Anaconda, 2026)
- Hands-On AI: Introduction to Retrieval-Augmented Generation (LinkedIn Learning, 2026)
- AI/ML Engineer Stage 1 (SLIIT, 2023)
- Artificial Intelligence Foundations: Machine Learning (LinkedIn Learning, 2026)
- Complete Guide to Generative AI for Data Analysis and Data Science (LinkedIn Learning, 2026)
- Machine Learning with Python: Foundations, Decision Trees, Logistic Regression, k-Means Clustering (LinkedIn Learning, 2026)
- Processing Text with Python Essential Training – NASBA CPE (LinkedIn Learning and NASBA, 2026)
- Certificate in Business Accounting (CIMA, 2023)